

Home Assignment 1

Min Zhu & Ziyi Wang

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Problem 1

(a) Compute Marginal Posteriors

By examining the three marginal posterior distributions required in the problem

$$f(\theta|\lambda, t, \tau), f(\lambda|\theta, t, \tau), f(t|\theta, \lambda, \tau)$$

we observe that τ consistently appears as a conditioning variable. This is consistent with the practical interpretation: given the observed disaster data τ , we aim to infer the distributions of θ , λ , and t . Therefore, we first write down the joint posterior distribution.

$$\begin{aligned} f(\theta, \lambda, t|\tau) &= \frac{f(\theta, \lambda, t, \tau)}{f(\tau)} \propto f(\theta, \lambda, t, \tau) \\ &\propto f(\tau|\lambda, \theta, t) \cdot f(\lambda|\theta, t) \cdot f(\theta|t) \cdot f(t) \end{aligned}$$

Furthermore, based on the modeling assumptions, the following conditional independence relationships hold: the observed disaster times τ depend only on λ and t , and are independent of θ , i.e. $f(\tau|\lambda, \theta, t) = f(\tau|\lambda, t)$; The intensity parameter λ depends only on θ , giving $f(\tau|\lambda, \theta, t) = f(\tau|\lambda, t)$; Moreover, θ is independent of t and follows its prior distribution, so $f(\theta|t) = f(\theta)$.

Finally, $f(\theta, \lambda, t|\tau) \propto f(\tau|\lambda, t) \cdot f(\lambda|\theta) \cdot f(\theta) \cdot f(t)$. The corresponding formulas are shown in Tabel 1.

Table 1: Formulas

LHS	RHS
$f(\tau \lambda, t)$	$\exp\left(-\sum_{i=1}^d \lambda_i(t_{i+1} - t_i)\right) \prod_{i=1}^d \lambda_i^{n_i(\tau)}$
$f(\lambda \theta)$	$\theta^{2d} \left(\prod_{i=1}^d \lambda_i\right) \exp\left(-\theta \sum_{i=1}^d \lambda_i\right)$
$f(\theta)$	$\theta e^{-\vartheta\theta}$
$f(t)$	$\prod_{i=1}^d (t_{i+1} - t_i) \cdot I_{\{t_1 < t_2 < \dots < t_{d+1}\}}$

1. $f(\theta|\lambda, t, \tau)$

$$\begin{aligned} f(\theta|\lambda, t, \tau) &\propto f(\lambda|\theta) \cdot f(\theta) \propto \theta^{2d} \left(\prod_{i=1}^d \lambda_i\right) \exp\left(-\theta \sum_{i=1}^d \lambda_i\right) \cdot \theta e^{-\vartheta\theta} \\ &\propto \theta^{2d+1} \cdot \exp\left(-\theta\left(\sum_{i=1}^d \lambda_i + \vartheta\right)\right) \end{aligned}$$

This expression follows a Gamma distribution of the form $\Gamma(2d + 2, \sum_{i=1}^d \lambda_i + \vartheta)$.

2. $f(\lambda|\theta, t, \tau)$

$$\begin{aligned} f(\lambda|\theta, t, \tau) &\propto f(\lambda|\theta) \cdot \exp\left(-\sum_{i=1}^d \lambda_i(t_{i+1} - t_i)\right) \prod_{i=1}^d \lambda_i^{n_i(\tau)} \\ &\propto \prod_{i=1}^d \lambda_i^{n_i(\tau)+1} \exp\left(-\sum_{i=1}^d \lambda_i(t_{i+1} - t_i + \theta)\right) \end{aligned}$$

This expression follows a Gamma distribution of the form $\Gamma(n_i(\tau) + 2, t_{i+1} - t_i + \theta)$.

3. $f(t|\theta, \lambda, \tau)$

$$\begin{aligned} f(t|\theta, \lambda, \tau) &\propto f(t) \cdot f(\tau|\lambda, t) \propto \prod_{i=1}^d (t_{i+1} - t_i) \cdot I_{\{t_1 < t_2 < \dots < t_{d+1}\}} \cdot \exp\left(-\sum_{i=1}^d \lambda_i(t_{i+1} - t_i)\right) \prod_{i=1}^d \lambda_i^{n_i(\tau)} \\ &\propto \exp\left(-\sum_{i=1}^d \lambda_i(t_{i+1} - t_i)\right) \prod_{i=1}^d (t_{i+1} - t_i) \cdot I_{\{t_1 < t_2 < \dots < t_{d+1}\}} \end{aligned}$$

This expression does not follow any standard known distribution.

(b) Construct hybrid MCMC

From the result in part (a) and the guidance provided in the problem, the marginal posterior distributions of both λ and θ follow Gamma distributions with known parameters. Therefore, Gibbs sampling is employed to update these two variables. For the breakpoints t , we instead apply the Metropolis–Hastings algorithm. Specifically, a random walk proposal is used due to its symmetry, under which the proposal distribution cancels out in the acceptance ratio. Each iteration k consists of the following three steps:

Step 1: Gibbs Update for θ . Draw the hyperparameter $\theta^{(k+1)}$ from its full conditional Gamma distribution:

$$\theta^{(k+1)} \sim \Gamma\left(2d + 2, \vartheta + \sum_{i=1}^d \lambda_i^{(k)}\right)$$

Step 2: Gibbs Update for λ . Count the number of disasters $n_i(\tau)$ falling within each sub-interval, then sample the intensities $\lambda_i^{(k+1)}$ independently for $i = 1, \dots, d$:

$$\lambda_i^{(k+1)} \sim \Gamma\left(n_i(\tau) + 2, t_{i+1}^{(k)} - t_i^{(k)} + \theta^{(k+1)}\right)$$

Step 3: MH Update for Breakpoints t . The boundaries are fixed at $t_1 = 1851$ and $t_{d+1} = 1963$. Each internal breakpoint t_j (for $j = 2, \dots, d$) is updated as follows:

- Proposal: Generate a candidate $t_j^* = t_j + \epsilon$, where $\epsilon \sim \text{Unif}(-R, R)$ and $R = \rho(t_{j+1} - t_{j-1})$.
- Constraint: Reject if the ordering constraint $t_{j-1} < t_j^* < t_{j+1}$ is violated.
- Acceptance Probability: When t_j is updated, only the two segments adjacent to it are affected. Therefore, $\ln \alpha = \min(0, \mathcal{L}_{\text{local}}(t_j^*) - \mathcal{L}_{\text{local}}(t_j))$ where

$$\mathcal{L}_{\text{local}}(t) = \sum_{m \in \{j-1, j\}} [n_m^* \ln \lambda_m - \lambda_m \Delta t_m + \ln \Delta t_m]$$

- Decision: Accept $t_j^{(k+1)} = t_j^*$ if $\ln u < \ln \alpha$ for $u \sim \text{Unif}(0, 1)$; otherwise, retain $t_j^{(k+1)} = t_j^{(k)}$.

(c) MCMC Behavior across Breakpoints

To investigate the impact of different numbers of breakpoints on MCMC performance, we consider $[1, 2, 3, 4]$ breakpoints, corresponding to $d = [2, 3, 4, 5]$. The results are illustrated in Fig. 1.

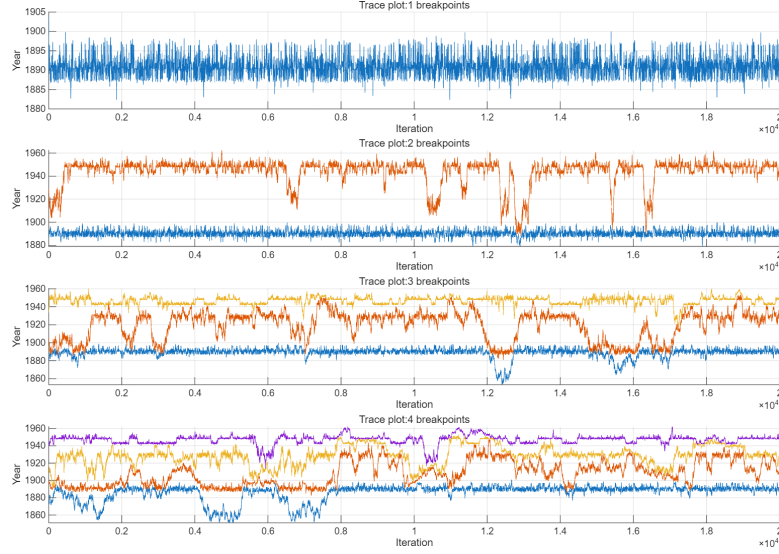
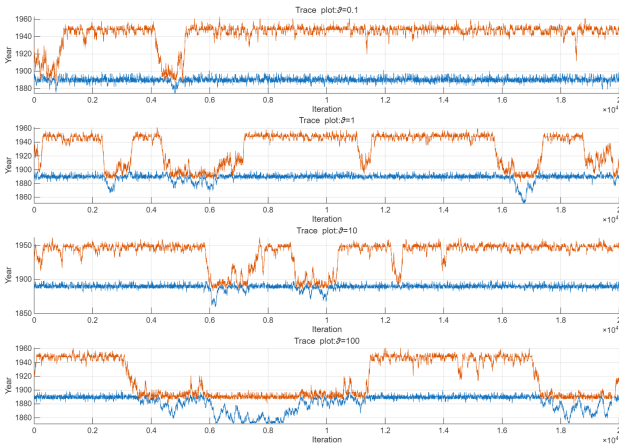


Figure 1: MCMC across breakpoints

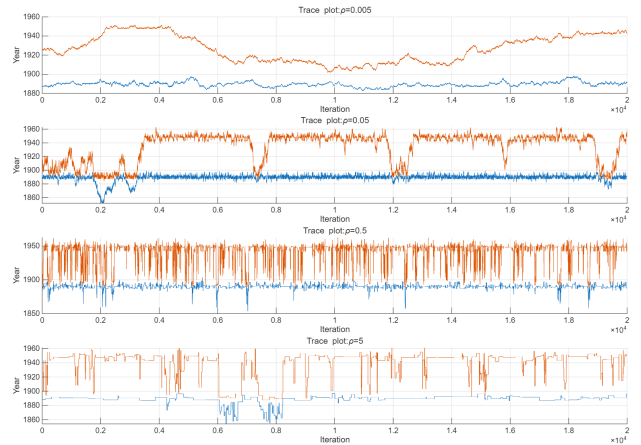
Observation

When a single breakpoint is used, the chain exhibits stable and orderly convergence around 1892. As the number of breakpoints increases to three and four, the variability grows, and the sampled trajectories display more pronounced fluctuations.

(d) MCMC Behavior across ϑ



(a) MCMC across ϑ



(b) MCMC across ρ

Figure 2: Sensitive Analysis

Observation

The hyperparameter ϑ encodes our prior belief about the intensity parameter before observing the data, through $\theta \sim \Gamma(2, \vartheta)$. As shown in Fig. 2a, varying ϑ across several orders of magnitude $[0.1, 1, 10, 100]$ leads to similar MCMC behavior. This indicates that the model is relatively insensitive to the choice of ϑ , suggesting that the posterior inference is largely driven by the data rather than the prior specification.

(e) MCMC Behavior across ρ

Observation

The parameter ρ controls the step size in the random walk proposal. The MCMC performance under different values of ρ is illustrated in Fig. 2b.

- $\rho = 0.005$: Under this very small step size, the trajectories of the breakpoints are highly smooth, continuous, and the fluctuations are tiny too. This is because the candidate values generated are extremely close to the current state. Consequently, the local posterior ratio is always near 1, leading to an exceptionally high acceptance rate.
- $\rho = 0.05$: This step size has best performance among all ρ . The blue chain stabilizes and exhibits small fluctuations around 1890, while the orange chain shows moderate variation around its overall trajectory. This step size is sufficiently large to allow efficient exploration of the target distribution, yet small enough to maintain a reasonable acceptance rate. As a result, the chain is able to generate approximately independent samples from the posterior distribution.
- $\rho = 0.5$: Under this step size, the vertical jumps become noticeably larger. This indicates that as ρ increases, the sampler attempts bolder moves. The acceptance rate begins to drop because many large jumps land in lower-density probability regions.
- $\rho = 5$: The trajectory exhibits a distinctive staircase pattern, characterized by prolonged horizontal lines punctuated by sudden, rare steps. This is because the massive step size means the proposal t_j^* frequently violates the boundary constraints ($t_{j-1} < t_j^* < t_{j+1}$) thus the acceptance rate is close to 0%, making the sampler highly inefficient.

Problem 2

(a) Derivation of the Log-Posterior and its Gradient

We are given conditionally independent observations modeled as $y_i | (\theta_1, \theta_2) \sim N(\theta_1^2 + \theta_2^2, \sigma^2)$ for $i = 1, \dots, n$. The parameter $\theta = (\theta_1, \theta_2)$ has a prior distribution $N_2(0, \Sigma)$

1. Likelihood Function

Given θ , the likelihood of the observations $y = (y_1, \dots, y_n)$ is:

$$L(\theta|y) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - (\theta_1^2 + \theta_2^2))^2\right)$$

2. Prior Density

The density function for the bivariate normal prior is:

$$f(\theta) \propto \exp\left(-\frac{1}{2}\theta^T \Sigma^{-1} \theta\right)$$

3. Log-Posterior $\ln f(\theta|y)$

According to Bayes' theorem, $f(\theta|y) \propto f(\theta)L(\theta|y)$. To compute the log-posterior $\ln f(\theta|y)$ up to an additive constant, we drop all terms that do not depend on θ :

$$\ln f(\theta|y) = -\frac{1}{2}\theta^T \Sigma^{-1} \theta - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (\theta_1^2 + \theta_2^2))^2 + c$$

where c is a constant independent of θ .

4. Gradient $\nabla_{\theta} \ln f(\theta|y)$

We need to compute the gradient $\nabla_{\theta} \ln f(\theta|y)$ with respect to θ . Let $S(\theta) = \theta_1^2 + \theta_2^2$. Taking the partial derivative with respect to θ_j (for $j = 1, 2$):

$$\frac{\partial}{\partial \theta_j} \ln f(\theta|y) = -(\Sigma^{-1}\theta)_j - \frac{1}{2\sigma^2} \sum_{i=1}^n 2(y_i - S(\theta)) \cdot (-2\theta_j)$$

Simplifying this yields the gradient vector:

$$\nabla_{\theta} \ln f(\theta|y) = -\Sigma^{-1}\theta + \frac{2}{\sigma^2} \left[\sum_{i=1}^n (y_i - (\theta_1^2 + \theta_2^2)) \right] \theta$$

(b) Hamiltonian Monte Carlo (HMC) Pseudocode

HMC introduces an auxiliary "momentum" variable $v \sim N(0, M)$, where the mass matrix M is typically set to the identity matrix I . The Hamiltonian is $H(\theta, v) = U(\theta) + K(v)$, where the potential energy is $U(\theta) = -\ln f(\theta|y)$ and the kinetic energy is $K(v) = \frac{1}{2}v^T v$. We use a leapfrog integrator to simulate Hamiltonian dynamics.

Pseudocode:

1. Initialize: Choose initial state $\theta^{(0)}$, step size ϵ , and number of leapfrog steps L .
2. Loop: For $m = 1, 2, \dots, N_{\text{samples}}$:
 - Sample initial momentum: $v^{(m-1)} \sim N(0, I)$
 - Set current state: $\theta^* = \theta^{(m-1)}$, $v^* = v^{(m-1)}$
 - Leapfrog Integration:
 - Half-step momentum update: $v^* \leftarrow v^* - \frac{\epsilon}{2} \nabla U(\theta^*)$
 - For $l = 1$ to $L - 1$:
 - * Full-step position update: $\theta^* \leftarrow \theta^* + \epsilon v^*$
 - * Full-step momentum update: $v^* \leftarrow v^* - \epsilon \nabla U(\theta^*)$
 - Final full-step position update: $\theta^* \leftarrow \theta^* + \epsilon v^*$
 - Final half-step momentum update: $v^* \leftarrow v^* - \frac{\epsilon}{2} \nabla U(\theta^*)$
 - Negate momentum for reversibility: $v^* \leftarrow -v^*$
 - Metropolis Accept/Reject Step:
 - Evaluate current Hamiltonian: $H_{\text{old}} = U(\theta^{(m-1)}) + \frac{1}{2}(v^{(m-1)})^T v^{(m-1)}$
 - Evaluate proposed Hamiltonian: $H_{\text{new}} = U(\theta^*) + \frac{1}{2}(v^*)^T v^*$
 - Calculate acceptance ratio: $\alpha = \min(1, \exp(H_{\text{old}} - H_{\text{new}}))$
 - Draw $u \sim U(0, 1)$. If $u < \alpha$, accept and set $\theta^{(m)} = \theta^*$; otherwise, reject and keep $\theta^{(m)} = \theta^{(m-1)}$.

(c) Implementation and Comparison

(i) Efficiency Analysis of Results

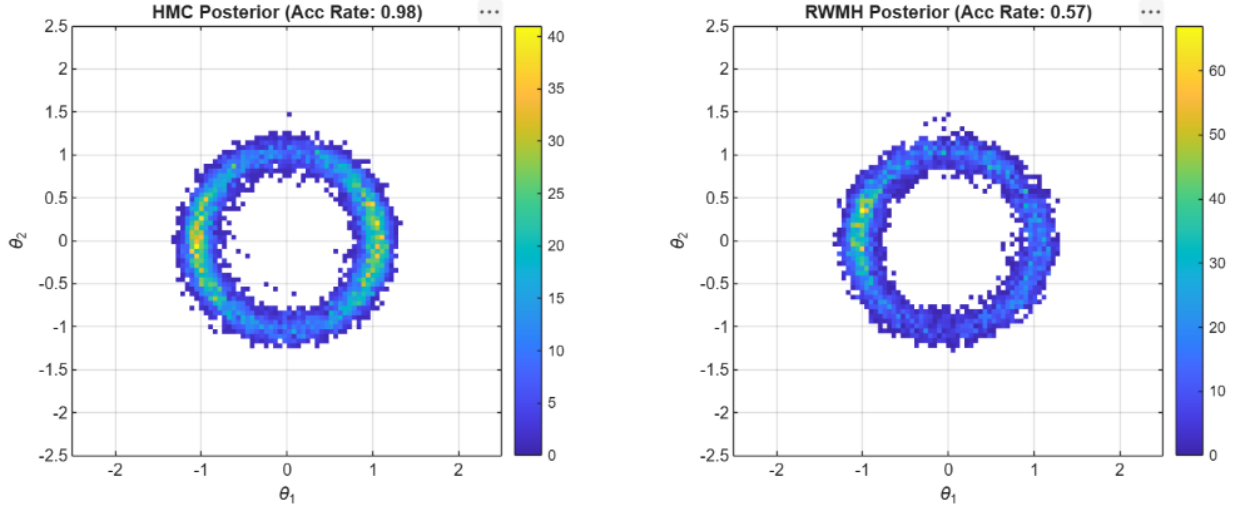


Figure 3: Posterior Density Estimation

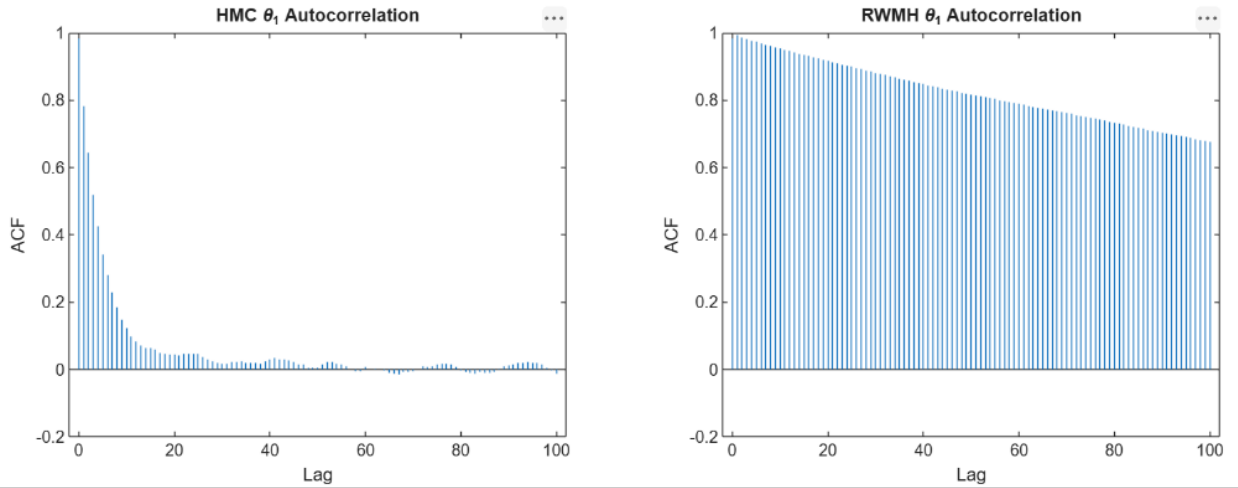


Figure 4: Autocorrelation (Mixing efficiency)

From Fig.3 and Fig.4, we observe a significant performance gap between the two methods:

1. Acceptance Rates: HMC achieves an extremely high acceptance rate of 98%, whereas RWMH only reaches 57%, under the chosen tuning parameters ($\epsilon = 0.05$, $L = 20$) and $\zeta = 0.15$. HMC maintains high acceptance because it uses the gradient to "glide" along the high-density ring, whereas RWMH makes blind jumps that often land in low-probability regions.
2. Support Exploration: The histogram for HMC shows a smooth, uniform ring, indicating it explores the entire parameter space effectively. The RWMH plot appears "clumpy," suggesting the sampler stays in local regions longer.
3. Autocorrelation (ACF): The HMC ACF drops to near zero with 30 lags, indicating highly independent samples and excellent mixing. In contrast, the RWMH ACF decays very slowly, remaining above 0.6 even at lag 100, which implies high redundancy and poor efficiency.

(ii) Parameter Tuning

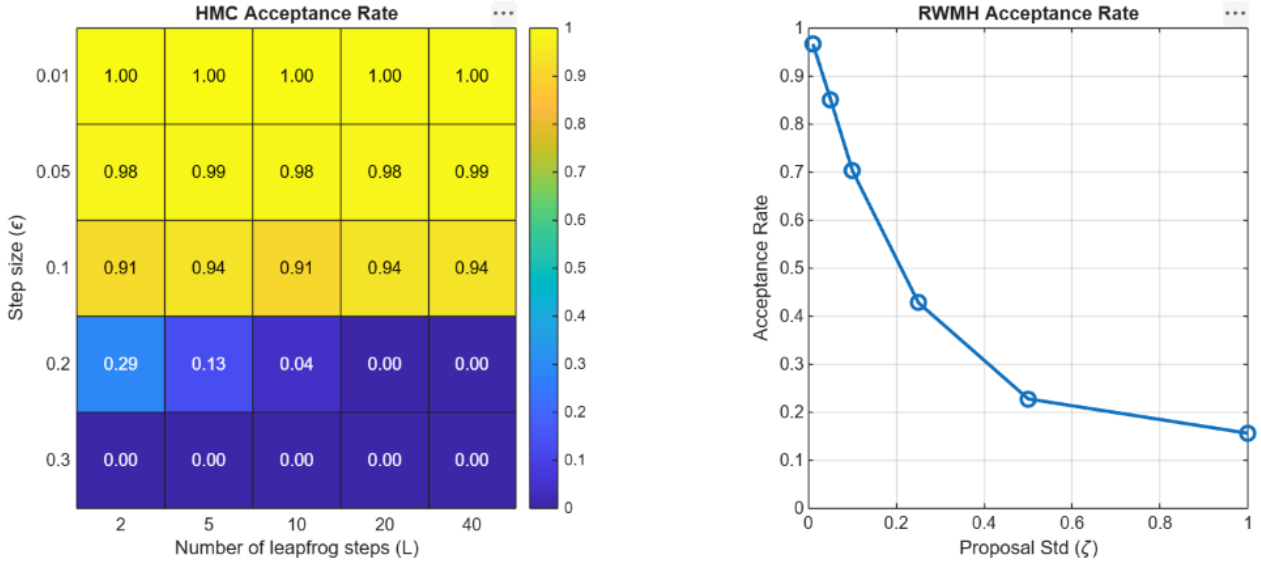


Figure 5: Parameter Tuning Results

1. HMC Performance Investigation (ϵ and L):

- **Sensitivity to Step Size (ϵ):** The algorithm is highly sensitive to the step size. For small values ($\epsilon = 0.01$ to 0.05), the acceptance rate remains nearly perfect at 0.98 to 1.00 regardless of the number of leapfrog steps. However, as ϵ increases to 0.2 , the acceptance rate drops significantly, and at $\epsilon = 0.3$, the algorithm fails to accept any samples (rate = 0.00).
- **Impact of Integration Steps (L):** At moderate step sizes (e.g., $\epsilon = 0.2$), increasing the number of leapfrog steps actually decreases the acceptance rate (from 0.29 at $L = 2$ to 0.00 at $L = 20$). This occurs because longer trajectories accumulate more numerical integration errors from the leapfrog method, leading to larger deviations in the Hamiltonian and a lower probability of acceptance.

2. RWMH Performance Investigation (ζ):

- **Inverse Relationship:** There is a clear inverse relationship between the proposal standard deviation (ζ) and the acceptance rate. When ζ is very small (near 0.01), the acceptance rate is nearly 1.00 . As ζ increases toward 1.0 , the acceptance rate decays rapidly to approximately 0.15 .

3. Comparative Summary

- **HMC Superiority:** Compared to RWMH, HMC allows for much larger "effective" steps through the parameter space while maintaining a significantly higher acceptance rate (near 1.00 for well-tuned ϵ).
- **Stability:** HMC remains robust over a wide range of L values as long as the step size ϵ is kept small enough to maintain integration accuracy. RWMH, however, requires careful calibration of ζ to balance the high acceptance of small steps against the better exploration of larger steps.